

Unit-3

MS-Excel:

- MS Excel is a general-purpose electronic spreadsheet used to organize, calculate and analyze data.
- Excel is a Microsoft Office Application.
- Microsoft Excel is the spreadsheet program.
- Excel can be used to create worksheets and workbooks with built-in mathematical functions.
- Excel is mainly used for making calculations and mathematical works.
- Microsoft Excel is a powerful spreadsheet software application developed by Microsoft.
- It is widely used for various tasks, such as data analysis, financial calculations, project management, and more.
- Excel provides a user-friendly interface and a range of features to work with Data efficiently.

Here are some key aspects of Microsoft Excel

- Alphabetical letters are usually assigned to columns and numbers are usually assigned to rows. The point where a column and a row meet is called a cell.
- The address of a cell is given by the letter representing the column and the number representing a row.

Features of MS excel

- **Spreadsheets:** Excel is primarily used for creating and managing spreadsheets, which are grids made up of rows and columns. Each intersection of a row and a column is called a cell, and it can contain data.
- **Formulas and Functions:** Excel allows users to perform calculations using formulas and functions. Formulas are user-defined expressions that perform calculations on values in a worksheet. Functions are predefined formulas that perform specific calculations.
- **Data Entry and Formatting:** Excel provides tools for data entry, formatting, and organization. You can input data into cells, format it to enhance readability, and use features like text and number formatting, cell merging, and conditional formatting.
- **Charts and Graphs:** Excel enables users to create a variety of charts and graphs to visually represent data. Common types of charts include bar charts, line charts, pie charts, and scatter plots. Excel provides customization options for these visualizations.
- **Data Sorting and Filtering:** Users can sort data alphabetically or numerically and filter data based on specific criteria. This is useful for organizing and analyzing large sets of information.

- **PivotTables:** PivotTables are powerful tools for summarizing and analyzing data. They allow users to reorganize and summarize selected columns and rows of data in a more readable format without altering the original data.

Use of MS EXCEL: -

- **Creating Spreadsheets:** Excel is commonly used to create spreadsheets for various purposes, such as tracking expenses, managing budgets, and maintaining lists.
- **Data Analysis:** You can use Excel to perform data analysis, including calculations, sorting, filtering, and creating pivot tables for more in-depth insights into your data.
- **Financial Modeling:** Excel is a popular tool for financial professionals and analysts for tasks like financial modeling, forecasting, and investment analysis.
- **Creating Charts and Graphs:** Excel's charting capabilities allow you to create visual representations of data, making it easier to understand and present information.
- **Project Management:** Excel can be used to create project schedules and Gantt charts, helping you manage tasks and deadlines.
- **Data Entry and Storage:** You can use Excel as a database to store and manage data, making it easy to search and retrieve information.
- **Creating Forms and Surveys:** Excel can be used to design forms and surveys, which can then be filled out by users and the data collected and analyzed.
- **Student Grading:** Teachers can use Excel to record and calculate student grades, simplifying the grading process.
- **Time Tracking:** Excel can be used to log hours worked on various projects, making it useful for time tracking and payroll.
- **Data Visualization:** You can use Excel to create interactive dashboards and visually appealing reports to communicate data insights effectively.
- **Personal Productivity:** Many people use Excel for personal tasks like creating to-do lists, tracking personal finances, and managing household expenses.
- **Goal Tracking:** Individuals and businesses can use Excel to set and track progress toward specific goals and objectives.
- **Address Books and Contact Lists:** Excel can be used to create and manage address books and contact lists, which can be useful for businesses and personal use.

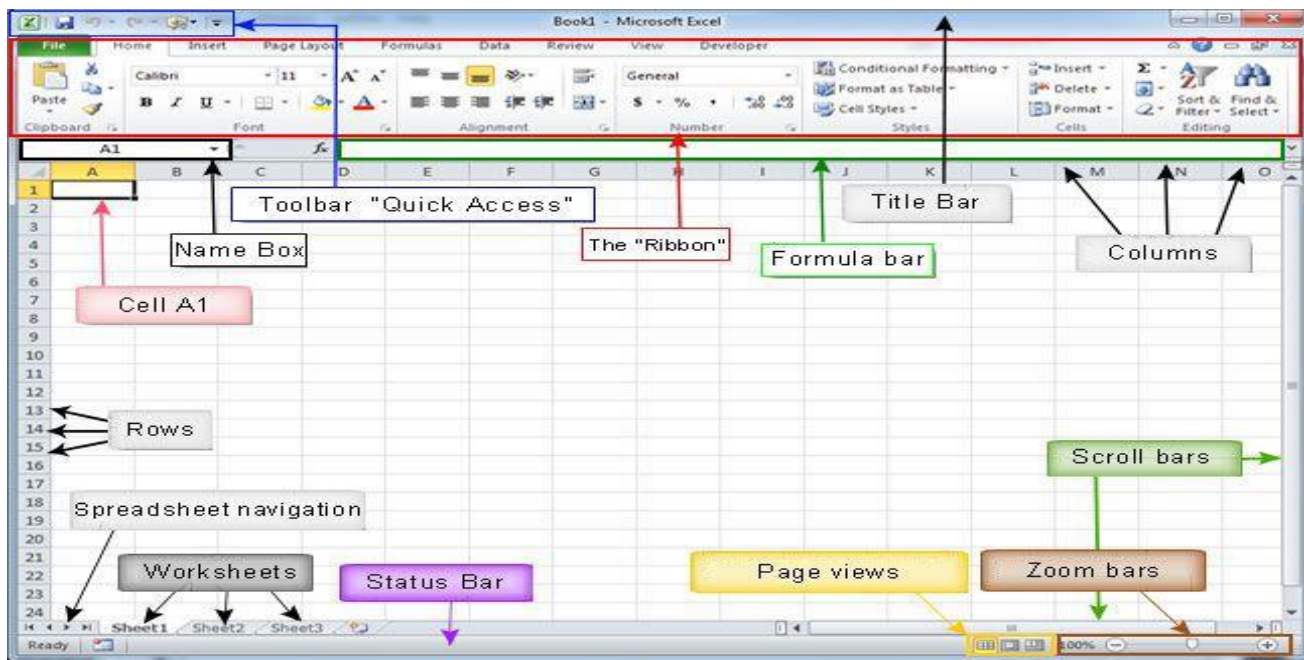
- **Statistical Analysis:** Researchers and statisticians use Excel for basic statistical analysis and graphing.

Three Major Parts of Excel are:

- a. **Worksheets:** Worksheets allow entering, calculating, manipulating and analyzing data such as numbers and text.
- b. **Charts:** Charts pictorially represent data. Excel can draw two-dimensional and three dimensional column charts, pie charts and other types of charts.
- c. **Databases:** Databases manage data. For example, once one enters data onto a worksheet, Excel can sort the data, search for specific data, and select data that meets certain criteria.

Different Parts of MS Excel Window:

- Title Bar
- Menu Bar
- Standard Toolbar
- Formatting Toolbar
- Cells
- Active Cell
- Row Heading and Column Heading
- Status Bar
- Scroll Bars
- Formula Bar



CELL IN MS EXCEL

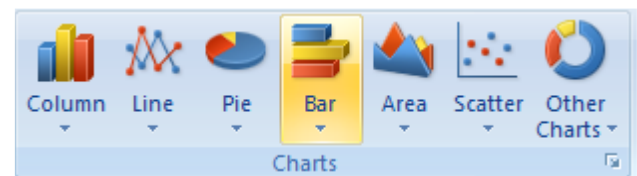
- In Microsoft Excel, a cell is the basic building block of a worksheet. It's the intersection point of a row and a column and is identified by a unique address, which is a combination of the column letter and row number. For example, the cell at the intersection of column B and row 3 is referred to as cell B3.
- Each cell in Excel can contain different types of data, such as text, numbers, formulas, or functions. You can enter and manipulate data within cells, perform calculations, format content, and apply various functions and formulas to analyze and work with the data.
- Excel cells are versatile and can be formatted in numerous ways, including changing the font, number format, background color, borders, and alignment to make the data more readable and visually appealing.
- Cells are fundamental in Excel as they are where users input, store, and manage data, forming the basis for creating spreadsheets and performing calculations or analysis.
- In Microsoft Excel, a cell address refers to the unique identifier of an individual cell within a worksheet. The cell address is a combination of the column letter and the row number, which specifies the exact location of a cell.

CELL ADDRESS IN MS EXCEL

- Each cell in Excel is identified by a cell address, which follows the standard notation of the column letter followed by the row number. For instance:
- The cell at the intersection of column B and row 3 is referred to as cell B3.
- The cell at the intersection of column F and row 10 is cell F10.
- **Cell Address Components:**
- **Column Letter:** Columns in Excel are labeled alphabetically. The first 26 columns are labeled A through Z. After reaching column Z, Excel continues with AA, AB, AC, and so on.
- **Row Number:** Rows are identified by their respective numbers, starting from 1 at the top of the worksheet.

Charts: -

- Charts are graphical representation of information.
- They are visually appealing and easy to understand and compare different patterns and trends in the data.
- Charts and Graphs are used to make information clearer and easier to understand
- In Microsoft Excel, creating charts is a powerful way to visually represent data. Excel provides various chart types that can help present data in a more understandable and compelling format.
- Charts is very important and useful function in Excel. In excel, we have different and good numbers of readymade chart options. We have multiple types of different charts in Excel Column, Bar, Radar, Line, Area, Combo, Pie and Bubbles chart.
- A chart is a visual representative of data in both columns and rows.
- Charts are usually used to analyze trends and patterns in data sets.



Types of Charts:

1)**Column chart**:-It is made up of horizontal or vertical column which represents the comparison of two or more things.

2) Line Chart:

•Line chart is represented through different data series and it represented through different colour of shading.

- It shows continuous change over the line.

3)Bar chart: -

A bar chart is a chart that shows information about two or more discrete objects, events, locations or groups of people, etc.

4)Pie Chart:

- It is a circular graph where the pieces of the pie are used to represent a percent (%)of a whole.
- These charts denote use horizontal and vertical axes to plot charts.
- The circle of the pie represents 100%, The size of each slice show which part of 100% it represents.

5) Area chart: -

- It emphasizes the magnitude of change over time by displaying the sum of plotted values, an area chart shows the relationship of parts to a whole.
- It combines some of the characteristics of a line chart with a bar chart.

6) Other chart: -

- Excel offers other chart types, such as Stock, Surface, Doughnut, Bubble, and Radar.
- To locate a menu of all available chart types in newer Excel versions, begin to insert any chart type and click**All Chart Types ...**

Steps to create a chart in MS Excel:

1. Open Excel
2. Enter the sample data
3. Select the data you want to represent in graph
4. Click on INSERT tab from the ribbon
5. Click on the Column chart drop down button
6. Select the chart type you want

Formula:

- It is user defined.
- It is an expression that is designed to perform calculation.
- It helps to calculate and analyze the data in the worksheet.
- Formulas can be used for simple calculation.

Formulas in Excel begin with an equal sign (=) and are used to perform calculations, manipulate data, and return specific values

Basic Properties of Formulas:

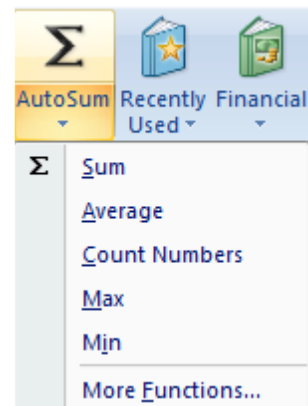
- A Formula in the worksheet always begins with an equal sign (=). This sign informs excel that formula is in use.
- After a formula is entered, the resulting value is displayed in the cell.
- When a cell containing a formula is selected, the underlying formula displays in the formula bar.
- **Operators:** Such as + (addition), - (subtraction), * (multiplication), / (division), ^ (exponentiation), etc.
- **References:** Cell references (e.g., A1, B3) or range references (e.g., A1:B10) to include data in calculations.

How To Use Formula In Ms Excel:-

- **Select a Cell:**
 - Click on the cell where you want the result of your formula to appear.
- **Start the Formula:**
 - Begin by typing an equal sign = in the selected cell. This signals that you're entering a formula.
- **Use Cell References or Values:**
 - Enter the formula using cell references (e.g., A1, B2, etc.), mathematical operators (+, -, *, /), functions, and/or constants to perform calculations.
- **Complete the Formula:**
 - Press Enter when you've finished entering the formula. The result will appear in the selected cell.

Function:

- It is a ready-to-use formula that can be used to perform a calculation on the data in the worksheet.
- **Functions in Excel** are pre-built formulas designed to carry out specific calculations or tasks.
- They are often more efficient than creating complex formulas from scratch. Excel provides a wide range of functions, including:
 - **Sum** : This Function is used to perform the addition of two or multiple operands.
 - **Average** : In order to take the average of more than one range using a single function, first start the = AVERAGE function. After adding your first range press "," (comma) and then select the second range of values, end the function with a parenthesis (bracket) and press
 - **MIN/MAX**
 - I. MIN Will return the smallest number found in a set of values.
 - II. MAX Will return the highest number found in a set of values.
 - **COUNT**
Counts the number of cells in a range than contain number
- **Mathematical Functions:** SUM, AVERAGE, MAX, MIN, ROUND, etc.
- **Statistical Functions:** STDEV, COUNT, CORREL, MEDIAN, etc.
- **Text Functions:** CONCATENATE, LEN, LEFT, RIGHT, UPPER, LOWER, etc.
- **Date and Time Functions:** TODAY, NOW, DATE, YEAR, MONTH, DAY, etc.
- **Logical Functions:** IF, AND, OR, NOT, etc.



Goal Seek

- In Microsoft Excel, the Goal Seek feature allows you to find the input value necessary to achieve a desired result in a formula.
- It's helpful when you want to reach a specific goal by changing the input variables.
- It's particularly useful when you have a specific target value in mind and need to adjust another value to achieve that goal.
- Goal Seek is a feature in Microsoft Excel that allows you to find the input value (such as a cell value) that will produce a desired result.

- It is useful when you have a target value in mind and want to determine what input is required to achieve that goal. Here's a step-by-step guide on how to use Goal Seek:

Using Goal Seek in Excel:

- **Open your Excel Spreadsheet:**

- Open the Excel file where you want to use Goal Seek.

- **Identify Your Goal:**

- Clearly identify the cell that contains the formula you want to change (the target cell), the cell with the desired result, and the cell you want to adjust to achieve the desired result.

- **Navigate to the "Data" Tab:**

- Go to the "Data" tab on the Excel ribbon.
- Locate the "What-If Analysis" Option:
- Within the "Data" tab, find the "What-If Analysis" option. This is where you'll find Goal Seek.

- **Select "Goal Seek":**

- Under "What-If Analysis," click on "Goal Seek."

- **Set Up the Goal Seek Dialog Box:**

- In the Goal Seek dialog box that appears, you will see three input fields:
- **Set cell:** This is the cell containing the formula you want to change (your target cell).
- **To value:** Enter the desired result that you want to achieve in this cell.
- **By changing cell:** This is the cell that Excel will adjust to meet the goal.
- **Configure Goal Seek:**
- Fill in the "Set cell," "To value," and "By changing cell" fields appropriately.
- Click "OK" when you've filled in the details.
- **Review the Results:**
- Excel will perform calculations to find the input value required to achieve the desired result. Once it's done, it will either provide a solution or inform you if it couldn't find a solution.
- Then okay close goal seeks.

PivotTable and Pivot Chart: -

- A PivotTable and a PivotChart are powerful features in Microsoft Excel that allow you to analyze and visualize data in a dynamic and interactive way.
- It is an interactive table that combines and compares large amounts of data very quickly.
- It creates an interactive summary from many records.
- One can rotate its rows and columns to see different summaries of the source data in order to produce meaningful information or details for areas of interest.

- A Pivot Table report is used to analyze related totals out of long list of figures to sum and to compare several facts about each figure.

Let's go through the process of creating a PivotTable and a PivotChart using a simple example.

Creating a PivotTable:

Prepare Your Data:

Ensure that your data is organized in columns with headers. Each column should represent a different attribute, and each row should contain a record.

Select Your Data:

Click anywhere in your dataset.

Insert a PivotTable:

Go to the "Insert" tab on the Excel ribbon.

Click on "PivotTable." This will open the "Create PivotTable" dialog box.

Choose Your Data Range:

The dialog box will automatically select the range based on your dataset. If not, manually enter the range.

Choose whether you want to place the PivotTable in a new worksheet or an existing one.

Design Your PivotTable:

In the PivotTable Fields pane, drag and drop the fields into the "Rows" and "Values" areas.

For example, you might drag a "Category" field to "Rows" and a "Sales" field to "Values" to see the total sales for each category.

Creating a PivotChart:

Create a PivotTable (if not done):

Before creating a PivotChart, it's often helpful to create a PivotTable first to summarize and organize your data.

Select Your PivotTable:

Click anywhere in your PivotTable.

Insert a PivotChart:

Go to the "Insert" tab on the Excel ribbon.

Click on "PivotChart." This will open the "Insert Chart" dialog box.

Choose Your Chart Type:

Select the type of chart you want to create (e.g., bar chart, line chart, pie chart).

You can preview how the chart will look in the right panel of the dialog box.

Design Your PivotChart:

In the "PivotChart Fields" pane, drag and drop fields to the "Axis," "Values," and "Legend" areas.

- By using PivotTables and Pivot Charts together, you can quickly analyze and visualize large datasets, making it easier to identify trends, patterns, and insights in your data.

Sorting Data in Excel:

Sorting data in Microsoft Excel allows you to organize information in a specific order based on the values in one or more columns.

- Sorting in Excel enables one to easily reorder the data based on the type of sorting required.
- One can sort data alphabetically or numerically in descending or ascending order or in any specific order.
- In Microsoft Excel, sorting data allows you to arrange the information in a worksheet either in ascending or descending order based on specific criteria.



- To sort in ascending order: **Alt + A + S + A**
- To sort in descending order: **Alt + A + S + D**
-

Here's a step-by-step guide on how to sort data in Excel:

Open Your Excel Spreadsheet:

Open the Excel file that contains the data you want to sort.

Select the Range:

Click on any cell within the column that you want to sort or select the entire range of data you want to sort.

Go to the "Data" Tab:

Navigate to the "Data" tab on the Excel ribbon.

Choose a Sorting Option:

There are different sorting options available:

Sort Ascending (A to Z): Click the "Sort A to Z" button (smallest to largest) for numerical or alphabetical sorting.

Sort Descending (Z to A): Click the "Sort Z to A" button (largest to smallest) for reverse sorting.

Specify Sorting Options:

In the "Sort" or "Custom Sort" dialog box, choose the column by which you want to sort.

Choose the sort order (ascending or descending).

If you're using the "Custom Sort" option, you can add additional sort levels and criteria.

Apply Sorting:

Click "OK" to apply the sorting options you've selected.

Data filtering: -

Data filtering in Microsoft Excel allows you to display only the information that meets specific criteria, making it easier to analyze and work with large datasets.

- Filtering is a way
 - To find and work a subset of data in a list.
 - To quickly extract certain data from the spreadsheet.
 - To set conditions so that only certain data is displayed.
- Filtering makes it easier to focus on specific information in a large database or table of data.
- Filters can be applied in many ways to improve the performance of the worksheet.
- MS Excel provides two commands for filtering lists:
 - **Autofilter:** It includes filter by selection, for simple criteria.

- **Advanced Filter:** It includes is for more complex criteria.

Here's a step-by-step guide on how to filter data in Excel:

Basic Filtering:

Open Your Excel Spreadsheet:

Open the Excel file containing the data you want to filter.

Select Your Data:

Click on any cell within the range of data you want to filter.

Go to the "Data" Tab:

Navigate to the "Data" tab on the Excel ribbon.

Apply AutoFilter:

Click the "Filter" button in the "Sort & Filter" group. This will add filter arrows to the column headers in your selected range.

Filter Data:

Click the filter arrow in the column header to open the filter options.

Unselect "Select All" and choose the specific values you want to display.

Click "OK" to apply the filter.

Clear Filters:

To clear filters and display all data again, click the filter arrow and select "Clear Filter."

Spreadsheet:

- It is a table of values arranged in rows and columns.
- Each value can have a predefined relationship to the other values.
- If one changes one value, therefore, one may need to change other values as well.
- Spreadsheet software is the type of application software used to create computerized spreadsheets, which typically contain a great deal of numbers and mathematical calculations.

Features of Spreadsheets:

- They perform calculations and recalculations automatically.
- They allow results to be displayed in tabular or graphic format.
- They consist of a number of worksheets.
- They consist of a table divided into rows and columns to produce cells.
- Spreadsheets allow to lay figures out on a grid, calculating and manipulating them visually.
- Spreadsheets software gives the ability to enter mathematical formulas ranging from simple arithmetic to complex statistics.
- Spreadsheets help in processing the information more quickly.
- Spreadsheet software includes a variety of data analysis tools, as well as ability to generate charts.

Data Validation:

- It is a tool that helps to control the kind of information that is entered in the worksheet.
- It helps to control the information that is entered in the worksheets.
- A method of validation is required to ensure that the data entered is correct.

Data Validation enables users to:

- Restrict entries to a specific type or size.
- Create custom settings.
- Create list of options / items in a cell.
- Set range of values that can be entered in a cell.
- Create a prompt message / warning explaining the kind of data allowed in a cell.

Data Validation is helpful in the following ways:

- One can avoid errors during the entry of data in the Excel cells.
- The data entry is quicker.
- The validation can be applied to text, numbers, time, date and custom data.
- One can provide hints during the data entry so that validation criteria are not involved.
- If wrong data is entered then the data validation process warns the operator through a message box.

Steps for Data Validation:

Select the Cell or Range:

Click on the cell or range of cells where you want to apply data validation.

Go to the "Data" Tab:

Navigate to the "Data" tab on the Excel ribbon.

Click on "Data Validation":

In the "Data Tools" group, click on "Data Validation."

Set Validation Criteria:

In the Data Validation dialog box, go to the "Settings" tab.

Choose the type of data validation you want (e.g., Whole Number, Decimal, List).

Configure Criteria:

Depending on the validation type, set the criteria and values.

Input Message (Optional):

Go to the "Input Message" tab to provide a message that will appear when the cell is selected.

Error Alert (Optional):

Go to the "Error Alert" tab to configure an error message that appears if an invalid entry is made.

Click "OK" to Apply:

Click "OK" to apply the data validation.

Conditional formatting

- Conditional formatting in Microsoft Excel is a feature that allows you to format cells based on specific conditions or criteria. This can help you visually analyze and interpret your data.
- Conditional formatting is a powerful tool that can significantly enhance the visual analysis of your data, making it easier to identify trends, patterns, and outliers.

- **Here's a step-by-step guide on how to apply conditional formatting in Excel:**

Basic Conditional Formatting:

- **Open Your Excel Spreadsheet:**

Open the Excel file containing the data you want to apply conditional formatting to.

Select Your Data:

- Highlight the range of cells you want to apply conditional formatting to.

Go to the "Home" Tab:

- Navigate to the "Home" tab on the Excel ribbon.

Access Conditional Formatting:

- In the "Styles" group, click on "Conditional Formatting."

Choose a Rule:

- From the dropdown menu, select the type of rule you want to apply. Common options include:

- **Highlight Cells Rules:** Apply formatting based on cell values (e.g., greater than, less than).
- **Top/Bottom Rules:** Highlight the top or bottom values.
- **Data Bars, Color Scales, and Icon Sets:** Apply visual formatting based on the values themselves.

Set Formatting Options:

- Configure the specific criteria and formatting options according to the selected rule.
- **Preview Changes:**
- Excel provides a live preview of how your formatting will look. Adjust the settings until you are satisfied.
- Click "OK" to Apply

- **linking and consolidation**

- In Microsoft Excel, linking and consolidation are two features that allow you to connect and combine data from multiple sheets or workbooks.
- These features are particularly useful when you need to analyze or summarize data from different sources. Let's take a closer look at each:
- **Linking Data Between Sheets:**
- Linking in Excel refers to creating a connection between cells or ranges in different sheets or workbooks. This connection allows changes in one location to be reflected in another automatically.
- **Consolidating Data in Excel:**
- Consolidation is the process of combining data from multiple sheets or workbooks into one summary sheet or workbook.
- This is useful when you have related information stored in different places and you want to analyze or present it collectively.
- The consolidation process can be done within a workbook or across different workbooks.
- These features in Excel provide flexibility and efficiency, allowing you to work with and analyze data from various sources in a cohesive manner. Linking and consolidation are particularly useful when dealing with large datasets or when collaborating with multiple contributors.